Raphaël Barboni

Postdoctoral fellow, Bocconi University

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Positions

2025 – present **Postdoctoral fellow**, *Bocconi University, Institute for Data Science and Analytics (BIDSA)*, Milan
mentored by **G. Zanella**.

Education

2022 – 2025 **Ph.D. in Mathematics**, *École Normale Supérieure – PSL, Department of Mathematics*, Paris
Thesis: "Training deep neural networks: mathematical models and convergence results",
supervised by **G. Peyré** and **F.-X. Vialard**.

2018 – 2022 **Élève fonctionnaire**, *École Normale Supérieure – PSL, Department of Mathematics*, Paris

2020 – 2021 **M.Sc. Mathematics for Machine Learning and Data Science (MVA)**, *ÉNS Paris-Saclay*, with honors
Thesis: *"Convergence properties of Gradient Descent in the training of Deep Residual Networks"*,
supervised by **G. Peyré** and **F.-X. Vialard**.

2018 – 2019 **B.Sc. in Mathematics**, *École Normale Supérieure – PSL*, Paris, with honors

Internships

2022 **Statistical to computational gaps in Tensor PCA**, *ETH, Mathematics Department*, Zürich
supervised by **A. Bandeira**

2021 **Convergence and implicit biases in training Deep Residual Networks**, *ÉNS (DMA)*, Paris
supervised by **G. Peyré** and **F.-X. Vialard**

2020 **Hydrodynamical models for red tide phenomena in Quellón's bay**, *Center for Mathematical Modeling (CMM) - CNRS*, Santiago, Chile
supervised by **C. Conca**

Teaching

2021 – 2025 **Teaching assistant**, *Université Paris Sciences et Lettres (PSL)*, Paris
"Cycle Pluridisciplinaire d'Études Supérieures" (CPES), 1st and 2nd year.

2018 – 2022 **Preparation for oral exams**, *Lycée Henri IV*, Paris

Publications

Published

R. B., Gabriel Peyré, and François-Xavier Vialard. "Understanding the training of infinitely deep and wide ResNets with conditional optimal transport". In: *Communications on Pure and Applied Mathematics* (2025).

R. B., Gabriel Peyré, and François-Xavier Vialard. "On global convergence of ResNets: From finite to infinite width using linear parameterization". In: *Advances in Neural Information Processing Systems* 35 (2022).

Preprints

R. B., Sebastiano Grazzi, and Giacomo Zanella. "Hamiltonian dynamics for sampling on discrete spaces". In: *arXiv preprint arXiv:2608.17961* (2026).

R. B., Maarten V. de Hoop, Takashi Furuya, and Gabriel Peyré. "Training Infinitely Deep and Wide Transformers". In: *arXiv preprint arXiv:2605.17660* (2026).

R. B., Gabriel Peyré, and François-Xavier Vialard. "Ultra-fast feature learning for the training of two-layer neural networks in the two-timescale regime". In: *arXiv preprint arXiv:2504.18208* (2025).

Conferences

Oral presentations

- SIAM - Mathematics of Data Science, November 2026, Salt Lake City, United States
- International Conference on Curves and Surfaces, June 2026, Saint-Malo, France
- Journées SMAI-MODE, March 2026, Nice, France
- International Workshop on Operator Theory (IWOTA), July 2025, Twente, Netherlands
- Signal, Image, Geometry, Modeling and Approximation (SIGMA), June 2024, Luminy, France
- PDE Methods in Machine Learning (Banff IRS event), June 2024, Granada, Spain

Poster presentations

- Foundations of Computational Mathematics (FoCM), July 2026, Vienna, Austria
- Scalable MCMC Sampling, June 2026, ETH Zürich, Switzerland,
- Physics of AI algorithms, January 2025, Les Houches, France
- Learning and Optimization in Luminy (LOL), June 2024, Luminy, France
- Workshop on Optimal Transport, from Theory to Applications, March 2024, Berlin, Germany
- Neural Information Processing Systems (NeurIPS), November 2022, New Orleans, USA
- International Conference on Curves and Surfaces, June 2022, Arcachon, France